# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 162 | Let breadth $= b$. |
| 2 | **(b)** $\frac{23}{24}$ | Use the LCM of the denominators. |
| 3 | **(c)** 2 | |
| 4 | **(d)** 5 | Number of sides $= 360 \div a$. |
| 5 | **(b)** 22 | Simplify step by step. |
| 6 | **(c)** 16 | Perimeters are equal. |
| 7 | **(b)** 60 | Difference of the fractions, then divide. |
| 8 | **(a)** 20 | Add the three sides. |
| 9 | **(b)** 45 | Divide. |
| 10 | **(d)** 30 | Opposites cancel. |
| 11 | **(d)** 120 | Half of the rectangle. |
| 12 | **(b)** 40 | Fraction removed = difference. |
| 13 | **(c)** 36 | Side $= 2r$. |
| 14 | **(c)** 2 | Letters H, I, O, X. |
| 15 | **(d)** 23 | Distance $= \lvert a - b \rvert$. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $l + b = 30$; ratio parts $= 5$ (1); Length $= 18$ cm, breadth $= 12$ cm (1). | 1+1 |
| 17 | Number $= 18 \div$ \frac{3}{4}$= 24$ (1); $\frac{1}{3}$ of the number $= 8$ (1). | 1+1 |
| 18 | Inside: 2; On: 1 (1); Outside: 1 (1). | 1+1 |
| 19 | Sides $= 360 \div 30 = 12$ (1); Perimeter $= 12 \times 4 = 48$ cm (1). | 1+1 |
| 20 | Opposite integers cancel each other; Remaining: integers from 7 to 10 (1); Sum $= 34$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Perimeter $= 130$ m; Wire needed $= 3 \times 130 = 390$ m (1); Fencing cost $= 4680$ (1); Ploughing cost $= 4000$ (1). | 1+1+1 |
| 22 | Girls $=$\frac{5}{8}$\times 480 = 300$ (1); Chess girls $=$\frac{2}{5}$\times 300 = 120$ (1); Boys $= 480 - 300 = 180$ (1). | 1+1+1 |
| 23 | Distance between adjacent centres $= 2r = 6$ cm (1); Side $= 6$ cm (1); Area $= 36$ cm$^2$ (1). | 1+1+1 |
| 24 | Angle $= 360 \div 5 = 72^\circ$ (1); Lines of symmetry $= 5$ (1); Total around centre $= 5 \times 72 = 360^\circ$ (1). | 1+1+1 |
| | **OR:** After dive: $-80$; After rise: $-40$ (1); After second dive: $-65$ (1); Total distance $= 115$ m (1). | 1+1+1 |
| 25 | Square perimeter $= 4 \times 12 = 48$ cm; rectangle perimeter $= 2(14 + 10) = 48$ cm; Square area $= 144$ cm$^2$ (1); Rectangle area $= 140$ cm$^2$ (1); Square is larger by $4$ cm$^2$ (1). | 1+1+1 |
| | **OR:** $\frac{?}{12} =$\frac{5}{6}$-$\frac{1}{4} (1); $= \frac{7}{12} =$\frac{7}{12} (1); Missing numerator $= 7$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Rectangle area $= 240$ cm$^2$ (1); Triangle area $= \frac{1}{2} \times 20 \times 8 = 80$ cm$^2$ (1); Remaining $= 160$ cm$^2$ (1); Fraction left $= \frac{160}{240}$ (2). | 1+1+1+2 |
| | **OR:** After day 1: \frac{2}{3} left (1); Day 2 sells $\frac{1}{6}$ of total (1); Left fraction $\frac{1}{2}$ (1); Total $= 30 \div \frac{1}{2} = 60$ m (2). | 1+1+1+2 |
| 27 | Draw AB = 6 cm (1); With A as centre and radius 10 cm draw an arc; mark the corner C using the perpendicular at B (1); Other side BC = 8 cm (1); Half diagonal OA $= 5$ cm (1); Perimeter $= 2(6 + 8) = 28$ cm (1). | 1+1+1+1+1 |
| | **OR:** (a) 6 (1); (b) $60^\circ$ (1); (c) 30 cm (1); (d) $240^\circ$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 336 | Outer minus inner. |
| 2 | **(b)** 8 | Divide the length by the piece length. |
| 3 | **(d)** 16 | Two diameters. |
| 4 | **(d)** 150 | Add the smallest angles. |
| 5 | **(a)** $-17$ | Subtract. |
| 6 | **(d)** 14 | Perimeter does not change. |
| 7 | **(c)** $5$ | Find the number first. |
| 8 | **(a)** 16 | Each is half the diagonal. |
| 9 | **(d)** 3 | If a figure has a line and order n, it has n lines. |
| 10 | **(c)** $-30$ | Gains positive, losses negative. |
| 11 | **(d)** 5 | Area = cost ÷ rate; breadth = area ÷ length. |
| 12 | **(c)** $\frac{7}{10}$ | Compare with a common denominator. |
| 13 | **(b)** 18 | Each side is $2r$. |
| 14 | **(a)** 3 | Order $n$ gives $n-1$ such turns before a full turn. |
| 15 | **(c)** $-6$ | The first two cancel. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Area = $1200$ m$^2$, perimeter = $140$ m; Levelling cost = $6000$ (1); Fencing cost = $1680$; total = $7680$ (1). | 1+1 |
| 17 | Required =\frac{5}{6}−\frac{2}{7} (1); = $\frac{23}{42}$ (1). | 1+1 |
| 18 | $\angle AOB = \angle COD = 35^\circ$ (vertically opposite) (1); $\angle BOC = \angle AOD = 145^\circ$ (1). | 1+1 |
| 19 | Order = $360 \div 40 = 9$ (1); $120^\circ$ is a multiple: 3 times 40 (1). | 1+1 |
| 20 | $? = 5 - (-9)$ (1); $? = 14$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Width of ring = $5$ cm (1); Inner diameter = $8$ cm (1); Longest chord of outer circle = its diameter = $18$ cm (1). | 1+1+1 |
| 22 | Lines through opposite vertices: $4 \div 2 = 2$ (1); Lines through midpoints of opposite sides: $2$ (1); Total = $4$ (1). | 1+1+1 |
| 23 | Other = $-4 - 11 = -15$ (1); Difference = $26$ (1); Check the sum: -4 (1). | 1+1+1 |
| 24 | Floor in cm: $600$ by $400$; Tile area $600$ cm$^2$ (1); Floor area $240000$ cm$^2$ (1); Tiles = $400$ (1). | 1+1+1 |
|  | **OR:** Together =\frac{1}{4}+\frac{1}{6}= $\frac{5}{12}$ (1); Remaining = $1 - \frac{5}{12} = \frac{7}{12}$ (1); State the answer (1). | 1+1+1 |
| 25 | BM = $5$ cm, BN = $3$ cm (1); Perimeter = $2(5+3) = 16$ cm (1); Area = $15$ cm$^2$ (1). | 1+1+1 |
|  | **OR:** Smallest angle = $360 \div 6 = 60^\circ$ (1); $180 \div 60 = 3$ rotations (1); A whole number means the figure fits (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Scores: -3, 3, -6, 6 (1); Final score $6$ (1); Lowest score $-6$ (1); Write all values as integers (2). | 1+1+1+2 |
|  | **OR:** Path parallel to length: $60 \times 3 = 180$ m$^2$ (1); Path parallel to breadth: $40 \times 3 = 120$ m$^2$ (1); Common square = $3 \times 3 = 9$ m$^2$ (1); Area of paths = $291$ m$^2$ (1); Rest of the park = $2109$ m$^2$ (1). | 1+1+1+1+1 |
| 27 | Sum of whole parts = $6$ (1); Sum of fractions = $1$ (1); Total = $7$ (1); After subtracting: $6\frac{1}{4}$ (2). | 1+1+1+2 |
|  | **OR:** Row length = $5 \times 6 = 30$ cm (1); First to last centres = $(5-1) \times 6 = 24$ cm (1); Circles in the strip = $7$ (1); Show the working with diameter (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 64 | A square has the greatest area for a given perimeter. |
| 2 | **(c)** $2\frac{3}{4}$ | Subtract. |
| 3 | **(c)** 70 | Base angles are equal. |
| 4 | **(b)** 9 | $360 \div$ angle. |
| 5 | **(b)** $-5$ | Add b to both sides. |
| 6 | **(b)** 42 | Add the three sides. |
| 7 | **(d)** 24 | LCM. |
| 8 | **(b)** 10 | Diameter equals the side. |
| 9 | **(b)** 42 | Number of sides times side. |
| 10 | **(b)** 2 | Subtracting a negative adds. |
| 11 | **(a)** 84 | Outer side $= a + 2$. |
| 12 | **(c)** $1\frac{1}{6}$ | Use the LCM of the denominators. |
| 13 | **(b)** 2 | |
| 14 | **(c)** 10 | Equal distance on the other side. |
| 15 | **(c)** 7 | All pairs cancel except the last. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Room area $= 48$ m$^2$; Carpet area $= 20$ m$^2$ (1); Uncovered $= 28$ m$^2$ (1). | 1+1 |
| 17 | One way $= \frac{23}{20}$ km (1); Total $= 2\times = \frac{23}{10}$ km (1). | 1+1 |
| 18 | Area of square $= 100$ cm$^2$ (1); Area of AOB $= 25$ cm$^2$ (one quarter) (1). | 1+1 |
| 19 | Smallest angle $= 360 \div 4 = 90^\circ$ (1); Angles: $90^\circ$, $180^\circ$, $270^\circ$ (1). | 1+1 |
| 20 | Final: $-20 + 60 - 25 = 15$ m (1); Distance from 30 m: 15 m (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Common difference $= 6$ (1); Next terms: 4, 10, 16 (1); First positive term: 4 (1). | 1+1+1 |
| 22 | Perimeter $= 160$ m; Wire needed $= 2 \times 160 = 320$ m (1); Fencing cost $= 4800$ (1); Ploughing cost $= 7500$ (1). | 1+1+1 |
| 23 | Girls $=\frac{4}{9}\times 360 = 160$ (1); Chess girls $=\frac{3}{8}\times 160 = 60$ (1); Boys $= 360 - 160 = 200$ (1). | 1+1+1 |
| 24 | Distance between adjacent centres $= 2r = 8$ cm (1); Side $= 8$ cm (1); Area $= 64$ cm$^2$ (1). | 1+1+1 |
| | **OR:** Angle $= 360 \div 6 = 60^\circ$ (1); Lines of symmetry $= 6$ (1); Total around centre $= 6 \times 60 = 360^\circ$ (1). | 1+1+1 |
| 25 | After dive: $-80$; After rise: $-25$ (1); After second dive: $-55$ (1); Total distance $= 155$ m (1). | 1+1+1 |
| | **OR:** Square perimeter $= 4 \times 15 = 60$ cm; rectangle perimeter $= 2(18 + 12) = 60$ cm; Square area $= 225$ cm$^2$ (1); Rectangle area $= 216$ cm$^2$ (1); Square is larger by 9 cm$^2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Angles: $120^\circ$ and $90^\circ$ (1); Difference $30^\circ$ (1); LCM of 120 and 90 is 360 (1); Common turn $360^\circ$ (2). | 1+1+1+2 |
| | **OR:** After down: $-2$ (1); After up: 0 (1); After down: $-3$ (1); Exists (2). | 1+1+1+2 |
| 27 | Rectangle area $= 240$ cm$^2$ (1); Triangle area $= \frac{1}{2} \times 24 \times 6 = 72$ cm$^2$ (1); Remaining $= 168$ cm$^2$ (1); Fraction left $= \frac{168}{240}$ (2). | 1+1+1+2 |
| | **OR:** After day 1: \frac{3}{4} left (1); Day 2 sells $\frac{1}{2}$ of total (1); Left fraction $\frac{1}{4}$ (1); Total $= 12 \div \frac{1}{4} = 48$ m (2). | 1+1+1+2 |

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